A Digital Twin for Shading-Aware Assessment of Vehicle-Integrated Photovoltaics on a Heavy Electric Vehicle

Aditya Kumar, Anandsagar Gaikwad, and Arjun Deo

***Abstract*—Feasibility studies of vehicle-integrated photovoltaics (VIPV) usually take the irradiance on a horizontal, unshaded roof. We extend SOLARIS, an hour-by-hour energy-balance toolkit for a $2.5\,\mathrm{t}$ seven-seat electric vehicle, with a digital twin of central Phoenix, Arizona. The twin builds a $0.5\,\mathrm{m}$ city model from 5.28 billion lidar points, drives the vehicle on traffic-simulated trips along three arterials, parks it at five sites and resolves the shadow on each facet of an assumed $11.6\,\mathrm{m}^2$ skin every second while driving and every five minutes while parked. On a satellite-derived typical year the conference model gives a solar self-sufficiency of 0.832 at $50\,\mathrm{km/day}$. Facet orientation lowers it to 0.678; with parking shade, street shade and electrical mismatch, a suburban commuter ends at 0.512 and a downtown street parker at 0.256. Most of the shading loss occurs while parked. We extend the conference design equation with orientation, shading and mismatch factors; it stays within 2.6% of the full simulation. A Unity twin agrees with the Python shading engine on 99.1% of 15.5 million sample-seconds. The pipeline and data recipes are open.**

***Index Terms*—Vehicle-integrated photovoltaics, digital twin, urban shading, electric vehicles, energy self-sufficiency.**

I. Introduction

SOLAR cells built into the body of an electric vehicle (EV) can supply part of the energy it would otherwise draw from the grid. How large that part is depends on sunlight, consumption and their timing. In an earlier paper [1] we built SOLARIS, an open toolkit that charges every harvesting component to the vehicle's mass, computes consumption on the WLTP cycle and steps the battery through a year hour by hour. For a $2.5\,\mathrm{t}$ seven-seat vehicle with an $11.6\,\mathrm{m}^2$ skin, satellite-derived weather gave a solar self-sufficiency f_{ss} of 0.83 in Phoenix, 0.71 in Lucknow and 0.52 in London for $50\,\mathrm{km/day}$ of urban driving.

Those numbers rest on two simplifications that most VIPV feasibility studies share. The whole skin receives global horizontal irradiance (GHI), although side panels are nearly vertical. Nothing shades the vehicle, although a car is parked for most of the day, often next to buildings or under trees, and drives through street canyons. Both make the result optimistic, and the size of the error depends on where the car is driven and parked.

This paper replaces both simplifications with a digital twin of central Phoenix and keeps the rest of the conference model unchanged, so each correction can be measured on its own. The contributions are:

1) a reproducible pipeline from public lidar, building and street data to a $0.5\,\mathrm{m}$ city model with buildings and trees separated, checked against survey marks and updated for buildings built or demolished since the lidar flight;
2) a facet-resolved irradiance, thermal and electrical model of the vehicle skin, driven by traffic-simulated trips at $1\,\mathrm{s}$ and by parking at $5\,\mathrm{min}$, with energy closure checked at every step;
3) an annual loss waterfall from the conference baseline to four daily-use archetypes, which separates the effects of facet orientation, parking shade, driving shade and electrical mismatch;
4) a shading-aware form of the conference design equation and a nomograph, checked against the full simulation;
5) a cross-check of the shading engines against an independent Unity ray-casting twin, an urban-climate model and street-level imagery.

II. Related Work

Reviews of VIPV agree that yield is set by geometry, curvature, shading and climate [3], [4]. Yield models compute harvest for fleets [5] or along shaded city routes [6] but take the vehicle's consumption as given; power-electronics studies optimize the converter and the auxiliary battery [7]. Urban shading has been estimated from geographic data: Brito *et al.* [8] found annual shading losses of up to 25% on Lisbon roads and over 50% at urban parking spaces, and Hoth *et al.* [9] report a median shading loss of 61% (quartiles 38% and 76%) for parked cars in Berlin. San José *et al.* [6] recorded shadows on a driving car with a camera and showed how partial shading of cells degrades the current–voltage curve. Sky view factors (SVF) for street routing in Phoenix have been computed from synthetic fisheye images [10], and the SOLWEIG model in UMEP [11], [12] is a standard reference for shadow patterns in urban climate work. To our knowledge no study has combined trajectory-level driving and parking

This work extends the paper "SOLARIS: A first-principles, drive-cycle- and climate-resolved methodology and open toolkit for assessing solar-assisted heavy electric vehicles," accepted for presentation at the IEEE Transportation Electrification Conference and Expo, Asia-Pacific (ITEC-AP 2026), Busan, Republic of Korea, 1–4 December 2026 [1].

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TABLE I
CONFERENCE BASELINE ON SATELLITE-DERIVED HOURLY TMY (TABLE B0)

City	Scen.	f_{ss}	Charges/yr (no solar)	Relief (%)	A_{req} (m^2)	$D_{\max}$ (km/day)
Phoenix	II	0.832	5 (27)	82	14.4	37.5
	I	0.419	31 (54)	42	28.8	33.6
Lucknow	II	0.713	8 (27)	71	16.9	29.6
	I	0.363	35 (53)	36	33.4	26.6
London	II	0.517	12 (24)	52	23.9	20.5
	I	0.248	39 (50)	25	50.2	18.7

Relief: harvest used (harvest minus curtailment) over demand. $D_{\max}$: distance that $11.6\,\mathrm{m}^2$ makes self-sufficient. TMY annual GHI 2110.0, 1843.6 and 1050.7 $\mathrm{kWh/m}^2$.

shade, facet orientation and electrical mismatch with drive-cycle demand and an annual battery simulation, or done so for a heavy vehicle.

III. CONFERENCE BASELINE

The conference model [1], [2] is used unchanged as the reference (version 1.1.0; the refactored code reproduces every output of the conference code, as the repository tests check). Demand is the WLTP consumption at the battery, $210\,\mathrm{Wh/km}$ on the city part of the cycle and $237\,\mathrm{Wh/km}$ on the full cycle, plus auxiliaries and cabin cooling. Harvest is

$$E_{\mathrm{pv}} = A\,\eta_{\mathrm{pv}}\,\kappa_{\mathrm{pv}}\,\frac{G}{1000}\,[1+\beta(T_c-25)] \tag{1}$$

with $A = 11.6\,\mathrm{m}^2$, $\eta_{\mathrm{pv}} = 0.22$, $\kappa_{\mathrm{pv}} = 0.80$, $\beta = -0.4\%/\mathrm{K}$, $T_c = T_a + 25G/1000$ and G the GHI, plus small thermoelectric and suspension terms. The $250\,\mathrm{kWh}$ pack is simulated hour by hour over a periodic year; it is recharged from 10% to 80% when needed. Scenario II is $50\,\mathrm{km/day}$ on the city cycle and Scenario I $100\,\mathrm{km/day}$ on the full cycle, and f_{ss} is annual harvest over annual demand. Setting harvest equal to demand gives the design area

$$A_{\mathrm{req}} = \frac{D\,c + E_{\mathrm{aux}}}{\eta_{\mathrm{pv}}\,\mathrm{PSH}\,\kappa_{\mathrm{pv}}\,f_T}, \tag{2}$$

where D is the daily distance, c the battery-side consumption, PSH the daily peak-sun hours, f_T the irradiance-weighted temperature factor and E_{aux} the daily auxiliary energy.

Table I is the baseline on satellite-derived hourly typical meteorological years (TMY): NSRDB for Phoenix and Lucknow [13] and PVGIS for London [14]. Everything that follows is for Phoenix, Scenario II unless stated, starting from $f_{\mathrm{ss}} = 0.832$, 5 grid charges per year (27 without solar) and $A_{\mathrm{req}} = 14.4\,\mathrm{m}^2$. The idealized synthetic climate of the conference paper gives 0.94; of the gap of 0.106, 0.058 comes from the annual irradiation and 0.047 from the timing of cloud.

IV. DIGITAL TWIN

Fig. 1 shows the twin. Python does all physics; Unity replays the same trips against the same geometry and is used only to check visibility (Section VIII). A Snakemake workflow builds every number and figure in this paper from the raw data, with fixed seeds.

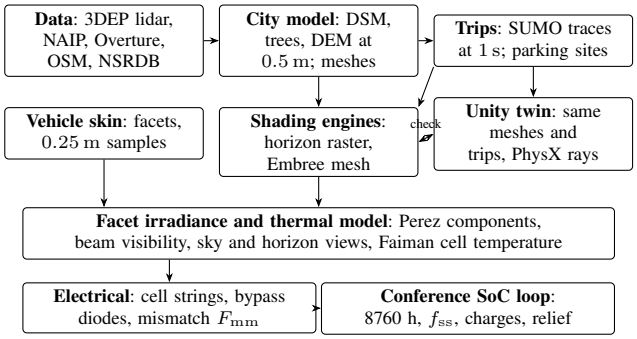


Fig. 1. Structure of the twin. Solid arrows carry data into the energy chain; the dashed link is the independent visibility check.

A. City Model

The area of interest is a $238\,\mathrm{km}^2$ box around the routes and parking sites (Fig. 2). USGS 3DEP lidar flown in 2020 [15] gives 5.28 billion points after noise removal, 22.2 points/m^2. Each return is binned into its own $0.5\,\mathrm{m}$ cell to form a surface model (DSM), a ground model (DEM) and a canopy model; 23.9% of the area is building and 4.4% tree. Buildings and trees are separated with 2021 NAIP imagery [16] and Overture building footprints [17]. Two corrections were needed. NAIP relief displacement paints tower walls green, so pixels taller than $35\,\mathrm{m}$ are never trees. Power lines, poles and signal arms formed thin walls along the arterials and shaded even the most open route at noon, so height-only structures narrower than $1.5\,\mathrm{m}$ are removed. Against 87 survey marks of the National Geodetic Survey the DEM has a median offset of −0.11 m (MAD 0.14 m). Of 159 buildings that differ between the lidar and current footprints along the routes, the largest 45 were checked against current imagery: 14 added (among them a 23-floor tower on the dense route), 12 removed and 19 rejected as bridges, decks or trees. OpenStreetMap [18] bridges and tunnels mark covered road. Delatin meshes of the DSM and the trees feed the mesh engine and Unity.

B. Routes, Traffic and Parking Sites

Routes were chosen by a fixed rule from 128 candidate $8\,\mathrm{km}$ windows on the named arterials: the highest mean SVF (R-OPEN, Southern Avenue, SVF 0.96), the lowest (R-DENSE, Central Avenue through downtown, 0.74) and the highest SVF variance in the middle band (R-MIXED, 19th Avenue, 0.90). SUMO [19] on an OpenStreetMap network produced 120 ego traces at $1\,\mathrm{Hz}$: both directions, five departures from 08:00 to 16:00 and three traffic demands. The five parking sites (Table II) span an open lot (P1), a curb in the downtown canyon (P2), a residential driveway beside a tree (P3), a garage roof deck (P4) and the P1 stall under a $4\,\mathrm{m}\times3\,\mathrm{m}$ shade roof (P5).

C. Shading Engines

The raster engine marches horizons through the DSM in 360 azimuths at 0.5° elevation steps out to $1\,\mathrm{km}$, once with trees transparent and once with trees opaque, and corrects azimuths for meridian convergence. Beam visibility of a sample is 1

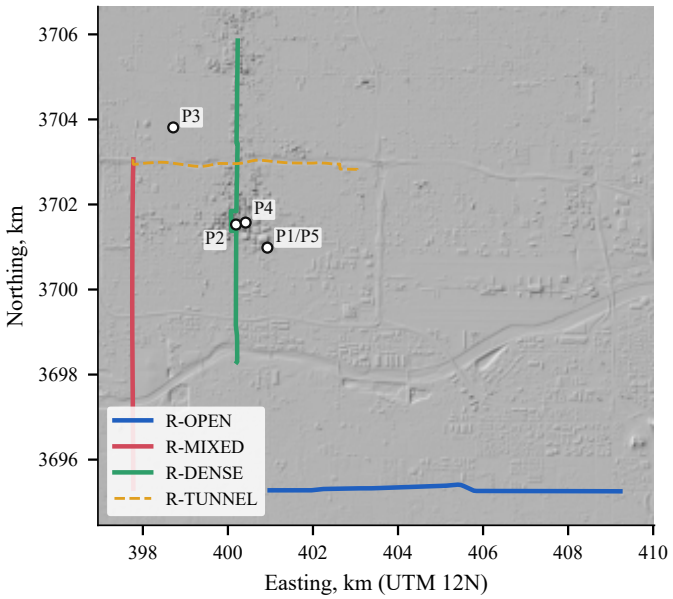


Fig. 2. Routes and parking sites over a hillshade of the DSM (UTM zone 12N). P5 is the P1 stall under a modelled shade roof.

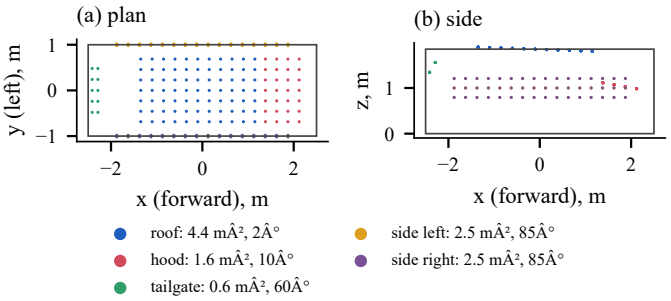


Fig. 3. Assumed design layout of the $11.6\,\mathrm{m}^2$ skin with the $0.25\,\mathrm{m}$ shading samples (area and tilt per facet).

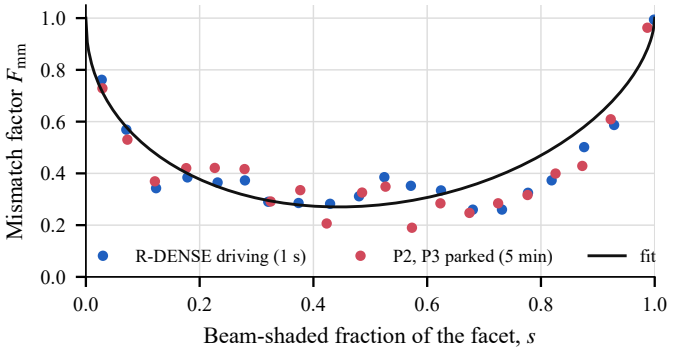


Fig. 4. Electrical mismatch factor against the beam-shaded fraction of a facet (energy-weighted bin means) and the fit (4).

when lit, 0 behind a building and $\tau = 0.03$ behind a tree. The mesh engine casts Embree rays [20] against the meshes. Sun positions come from the NREL SPA algorithm [21] in pvlib [22]. The two engines and SOLWEIG were verified in 40 checks covering sun position, transposition, shadow length and canyon SVF; all passed (Table V).

V. Vehicle Skin and Energy Chain

The intended panel layout of the vehicle is not fixed, so we use an assumed design layout (Fig. 3) that totals the conference $11.6\,\mathrm{m}^2$: roof $4.4\,\mathrm{m}^2$ at 2°, hood $1.6\,\mathrm{m}^2$ at 10°, tailgate $0.6\,\mathrm{m}^2$ at 60° and two sides of $2.5\,\mathrm{m}^2$ at 85°. Like the mass audit, it is a design estimate; the roof alone is reported as a bound. Each facet is sampled on a $0.25\,\mathrm{m}$ grid. With the Perez model [23] split into beam (with circumsolar) G_b, isotropic G_d and horizon G_h components and ground reflection G_g (albedo 0.2), facet f receives

$$G_f = \bar{v}_{b,f}\,G_{b,f} + \mathrm{SVF}_f\,G_{d,f} + \bar{v}_{h,f}\,G_{h,f} + G_{g,f}, \qquad (3)$$

where $\bar{v}_{b,f}$ is the mean beam visibility of its samples, SVF_f the cosine-weighted sky view and $\bar{v}_{h,f}$ the view of the horizon band; all three are 1 in open sky and the sky terms are 0 on covered road. Cell temperature follows Faiman [24], $T_c = T_a + G_f/(u_0 + u_1 w)$ with pvlib's open-rack coefficients and w the wind plus the vehicle speed, and energy follows (1) with G_f for G.

Partial shade costs more electrically than its area suggests. Each facet is one module on its own MPPT input, its samples are cells, and the cells form three bypass-diode substrings. The current–voltage curves come from pvlib's implementation of Bishop's model [25] with reverse breakdown, using area-scaled parameters of a 22.1%-efficient back-contact cell. Under uniform light the string reproduces the averaged model within 7×10^{-5}. Over all R-DENSE trips and the P2 and P3 parking steps, the mismatch factor $F_{\mathrm{mm}} = P_{\mathrm{el}}/P_{\mathrm{avg}}$ is fitted against the beam-shaded fraction s of the facet (Fig. 4):

$$F_{\mathrm{mm}} = 1 - 1.49\,s^{0.46}(1-s)^{0.58}, \qquad R^2 = 0.84. \qquad (4)$$

One shaded cell already pushes its substring through the bypass diode, so F_{mm} drops steeply at small s.

Four daily-use archetypes keep the demand of their baseline scenario and change only where the car is when the sun shines. A1, a suburban commuter, drives R-MIXED and R-OPEN at 08:00 and 17:00 and parks at the open lot P1 by day and the driveway P3 at night. A2 and A3 drive R-DENSE and R-OPEN at the same hours and park downtown, at the canyon curb P2 (A2) or under the shade roof P5 (A3). A4, a fleet vehicle, drives $100\,\mathrm{km/day}$ on all three routes at 08:00 to 16:00 and alternates P2 and P1 between trips. Weekend daytime parking moves to the night site. Parked energy averages the headings that fit each site. Driving at $1\,\mathrm{s}$ and parking at $5\,\mathrm{min}$ are binned into the 8760 h of the conference loop; binned and fine energies agree to 1×10^{-14} and driving plus parked seconds make 3600 in every hour.

VI. Results

A. Facet Orientation

Laid flat through the new chain, the skin reproduces the conference harvest within −0.03% (target ±0.5%); Faiman cooling adds 0.55%. Averaged over headings, the roof and hood see 1.00 and 0.99 of GHI, the tailgate 0.74 and the sides 0.56. Facet orientation alone lowers f_{ss} from 0.832 to 0.678

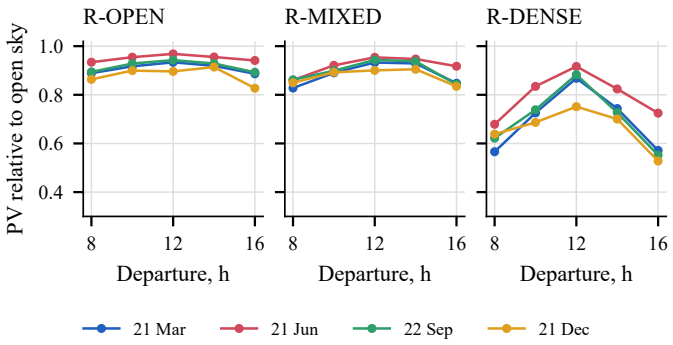


Fig. 5. Driving PV energy relative to open sky by departure time and date (buildings and trees, TMY, mean of both directions and three traffic demands).

TABLE II
ROUTES AND PARKING SITES: PV ENERGY RELATIVE TO OPEN SKY

Route	Length (km)	SVF	Buildings B/A	+ trees C/A	Beam shaded
R-OPEN	8.3	0.96	0.93	0.91	10%
R-MIXED	7.8	0.90	0.90	0.89	12%
R-DENSE	7.8	0.74	0.72	0.71	32%

Site	Setting	SVF	F_{park}
P1	open lot	0.95	0.94
P2	canyon curb	0.34	0.32
P3	driveway, tree	0.48	0.45
P4	garage roof deck	0.91	0.88
P5	P1 under shade roof	—	0.64

Driving: 360 trips on the TMY days (four dates, both directions, five departures, three demands), trees at $\tau = 0.03$. Parking: annual, heading-averaged.

($F_{\mathrm{geo}} = 0.805$) and raises grid charges from 5 to 9 per year. The roof alone would give 0.335.

B. Driving and Parking Shade

Table II lists the shading factors of the 360 trips and five sites on the TMY. While driving, buildings leave 0.93 of the open-sky harvest on R-OPEN and 0.72 on R-DENSE, and street trees take slightly more. Shading is strongest at the 08:00 and 16:00 departures, when the sun is low, and on R-DENSE the December noon trips also lose far more than the noon trips of the other dates (Fig. 5). Parking spans a much wider range: annual F_{park} is 0.94 at the open lot, 0.45 beside the tree and 0.32 at the canyon curb, and the shade roof leaves 0.64. Against SOLWEIG the parked shadow maps agree on at least 96.7% of pixels. The driving losses fall within the 25% that Brito *et al.* [8] found for Lisbon roads, and the curb and driveway losses lie between the Berlin quartiles of Hoth *et al.* [9].

C. Annual Loss Waterfall

Fig. 6 and Table III carry the baseline to each archetype one correction at a time. The archetype trip hours and the Perez–Faiman chain raise f_{ss} slightly (A1: 0.832 to 0.835). Facets then remove 0.159 for the 50 km archetypes. Parking shade is the largest and most variable step: 0.125 for the commuter, 0.256 under the shade roof and 0.369 at the canyon

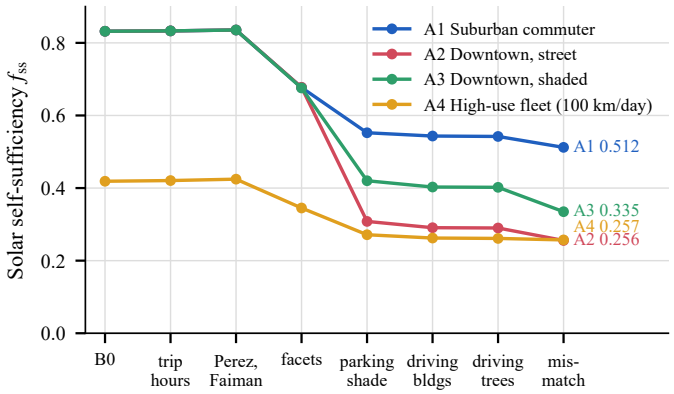


Fig. 6. Annual f_{ss} from the baseline (B0) to each archetype, one correction at a time (Phoenix, TMY).

TABLE III
ANNUAL RESULTS PER ARCHETYPE (PHOENIX, TMY)

	f_{ss}		Loss	Charges/yr	Relief	Grid
Archetype	B0	final	(%)	B0→final	(%)	(kWh/yr)
A1 commuter	0.832	0.512	38	5→13	51	2,300
A2 street	0.832	0.256	69	5→20	26	3,540
A3 shade roof	0.832	0.335	60	5→18	33	3,172
A4 fleet	0.419	0.257	39	31→41	26	7,251

A1–A3: 50 km/day, city cycle; A4: 100 km/day, full cycle. Loss: share of the B0 f_{ss}.

curb. Driving past buildings costs 0.009 to 0.017, and street trees 0.001 or less. Electrical mismatch costs 0.030 to 0.067. Energy-weighted, F_{mm} is 0.96 on the R-DENSE trips and 0.96 at the curb, where whole facets fall into building shadow, but 0.56 to 0.73 in the dappled shade of the tree at P3. In total the archetypes lose 38% (A1) to 69% (A2) of the baseline f_{ss}, and grid charges rise from 5 to between 13 and 20 a year. Replacing the hourly TMY with 2023 weather at 5 min changes final f_{ss} by at most 0.004, so hourly weather suffices for annual totals.

VII. SHADING-AWARE DESIGN INVERSION

The waterfall suggests three factors that the conference design equation lacks. From the annual harvest of each archetype at successive steps, $F_{\mathrm{geo}} = E_{\mathrm{facets}}/E_{\mathrm{flat}}$, $F_{\mathrm{sh}} = E_{\mathrm{shaded}}/E_{\mathrm{facets}}$ (parking and driving, weighted by where the car actually is) and $F_{\mathrm{mm}} = E_{\mathrm{final}}/E_{\mathrm{shaded}}$, and

$$A_{\mathrm{req}} = \frac{D\,c + E_{\mathrm{aux}}}{\eta_{\mathrm{pv}}\,\mathrm{PSH}\,\kappa_{\mathrm{pv}}\,f_T\,F_{\mathrm{geo}}\,F_{\mathrm{sh}}\,F_{\mathrm{mm}}}. \tag{5}$$

To test it, the per-facet harvest of each archetype was recomputed and scaled to other areas and allocations (roof only, roof and hood, full skin, with the area spread over the allocated facets in proportion), and the hour-by-hour simulation, with the vehicle mass fed back at 3.5 kg/m^2, was solved for the area at which $f_{\mathrm{ss}} = 1$. Over 60 cases (four archetypes, three allocations, $\kappa_{\mathrm{pv}} = 0.60$ to 0.80) (5) stays within 2.6% of the simulation (3.3% with the mass held fixed), inside the 3% target. Table IV lists the factors for the full skin and Fig. 7 the resulting design curves.

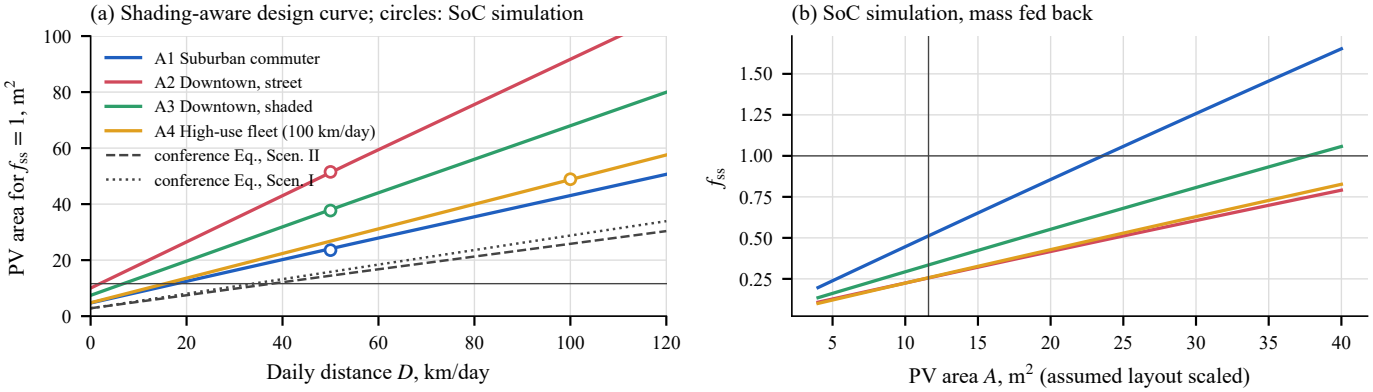


Fig. 7. (a) Area for $f_{ss} = 1$ against daily distance from (5) for each archetype (full skin, $\kappa_{pv} = 0.80$), with the hour-by-hour simulation (circles) and the conference equation (2) (gray). (b) f_{ss} against area in the simulation.

TABLE IV
LOSS FACTORS AND REQUIRED AREA (FULL SKIN, $\kappa_{pv} = 0.80$)

	F_{geo}	F_{sh}	F_{mm}	A_{req} (m^2)		Error	D_{max}
				(5)	sim.	(%)	(km/d)
A1	0.80	0.79	0.94	24.1	23.6	2.1	17.8
A2	0.80	0.40	0.87	51.3	51.5	−0.5	1.9
A3	0.80	0.58	0.82	38.0	37.7	0.8	6.8
A4	0.81	0.74	0.98	48.8	48.9	−0.3	15.5

B0 for comparison: $A_{req} = 14.4$ (Scen. II) and 28.8 m^2 (Scen. I). D_{max} at 11.6 m^2.

On the full skin, a commuter would need 24.1 m^2 instead of 14.4 m^2, and a downtown street parker 51.3 m^2, more than the roof, hood and sides of the vehicle can hold. Put the other way, the 11.6 m^2 skin covers 17.8 km/day for the commuter against 37.5 km/day in the baseline. A roof-only skin of the same area would have a higher F_{geo} but does not fit on the roof; the sweep table in the repository gives f_{ss} for every area, allocation, κ_{pv} and archetype.

VIII. VALIDATION OF THE TWIN

Without measured on-vehicle irradiance, the twin is cross-verified rather than physically validated (Table V, Fig. 8). Unity 6 with Cesium for Unity imported the same meshes around R-DENSE, replayed all 120 trip-dates and cast PhysX rays from every sample once per second. Its surface lies within 0.14 m of the lidar at 5 survey marks. Unity and the raster engine agree on 99.1% of 15.5 million sample-seconds; trip lit fractions agree within 2 percentage points (pp) on 119 of 120 trips (largest 2.37 pp). Embree on the same meshes reproduces Unity to 0.02 pp, which locates the remaining difference in the surface representation: the raster treats each pixel as a flat-topped column, while the mesh rounds roof edges. The $\pm 1^\circ$ band around the horizon line holds 3.2% of sample-seconds but 65% of disagreements; outside it every trip agrees within 0.65 pp.

Mapillary panoramas [26] within 10 m of R-DENSE and R-MIXED and at least 250 m apart give 8 test points. Their sky segmentation gives SVF with the engine's cosine weighting. Of these, 7 agree with MAE 0.054 and $r = 0.92$; the remaining

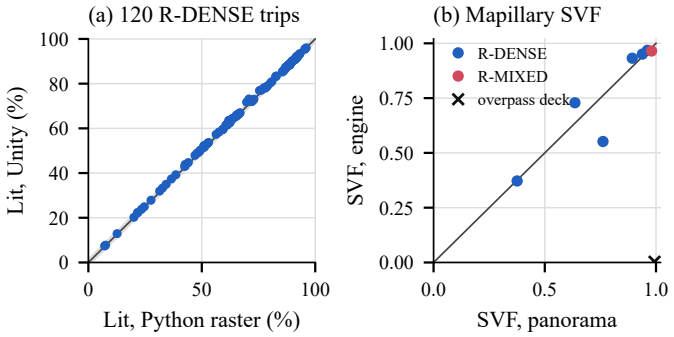


Fig. 8. (a) Trip lit fraction, Unity against the raster engine (band ±2 pp). (b) SVF from Mapillary panoramas against the engine at 2 m.

TABLE V
VERIFICATION AND VALIDATION METRICS

Check	Result	Target
Engine checks (sun, transposition, shadows, SVF)	40/40	all
Raster vs. mesh beam, 1000 points, 43 suns	99.6%	$\geq 98\%$
Raster vs. mesh SVF, mean $\lvert\Delta\rvert$	0.008	< 0.02
SOLWEIG shadow pixels, 12 times	98.0–99.3%	$\geq 95\%$
SOLWEIG at parking sites	$\geq$96.7%	$\geq 95\%$
Unity vs. raster, sample-seconds	99.1%	$\geq 97\%$
Unity vs. raster, trips within 2 pp	119/120	all
Mapillary SVF, MAE ($n = 7$)	0.054	reported
Energy closure, all steps	1×10^{-14}	$\leq 10^{-3}$

panorama was taken on an overpass deck that the route passes beneath, where the engine correctly sees the deck (all 8: MAE 0.171). With so few points this is a plausibility check.

IX. DISCUSSION AND LIMITATIONS

Where the car parks matters more than where it drives: the commuter keeps 0.512 because it parks in the open by day, while the same vehicle with the same trips keeps 0.256 at a downtown curb. For design, (5) turns a site survey into three multiplicative factors that can be estimated from shade maps without a full simulation, and the error of the closed form is smaller than the spread between archetypes.

[Figure placeholder: Unity render]
R-DENSE northbound, 21 December, 08:00 departure, medium traffic, analysis geometry in shadows-only mode under Google Photorealistic 3D Tiles. Scene as specified in PAPER_CHANGELOG.md (P8). Rendered by the authors.

Fig. 9. The Unity twin on R-DENSE on a winter morning, with the shadow on the skin from the analysis geometry. Imagery: Google; Cesium.

The method transfers to other cities. London's PVGIS weather is already in Table I; the geometry needs national lidar (the Environment Agency survey in England) and the same building and street sources. A lower sun and narrower streets would likely raise the parking and orientation losses there, so the London baseline of 0.517 should be read as an upper bound.

The limitations are as follows. The panel layout is an assumed design; side panels dominate F_{geo}, so a roof-heavy design would lose less to orientation. Mismatch depends on the module layout (three diodes per facet module and $0.25\,\mathrm{m}$ cells); more diodes or per-substring tracking would reduce it. Trees use one transmissivity and the lidar canopy of 2020, without seasonal leaf change. Traffic is simulated and the arterials stay below saturation, so queues in shade are underrepresented. The weather is a typical year, the shading and SVF were verified against models and imagery rather than measured irradiance on a vehicle, and 114 smaller building changes along the routes were not reviewed.

X. Conclusion

A digital twin of central Phoenix shows that the usual horizontal, unshaded roof overstates the solar self-sufficiency of a heavy VIPV vehicle: the four archetypes lose 38% to 69% of the baseline value. Starting from 0.832 at $50\,\mathrm{km/day}$, facet orientation and shade leave 0.512 for a suburban commuter and 0.256 for a car parked at a downtown curb, and grid charges rise from 5 to 13–20 a year. Parking shade is the largest loss after orientation; driving through street canyons costs at most 0.017. A shading-aware design equation reproduces the full simulation within 2.6% and gives the required area for a new site from three factors.

Code and Data Availability

The pipeline, configuration, frozen route manifest and result tables are at https://github.com/jarvisxys/solaris-vipv; the conference toolkit is at https://github.com/Anandsagar18/Solaris [2]. Lidar, NAIP and NSRDB data are public and fetched by the pipeline; Overture and OpenStreetMap data are under ODbL and Mapillary imagery under CC BY-SA 4.0.

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